

Bidhan Manna

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EDUCATION

Vellore Institute of Technology Aug 2022 – May 2026

Bachelor of Technology, Major in Computer Science and Engineering | Cumulative GPA: 7.86

TECHNOLOGIES AND TOOLS

Languages: Java, C#, Python, HTML/CSS, SQL **Tools:** Git, GitHub, MS Excel, Figma **Frameworks:** REST API, DBMS **Databases:** MongoDB, MySQL **Game Engine:** Unity 3D

CERTIFICATIONS

Java Foundation Associate, Oracle (Aug 2025) **Cyber Security Analyst**, IBM (May 2025) **Intro to AR/VR/MR/XR**, Univ. of Michigan / Coursera (Dec 2023) **Intro to Cloud Computing**, Simplilearn (Apr 2025)

WORK EXPERIENCE

Unity Developer Intern – iTeach (Remote/On-site) Mar 2026 – May 2026

- Designed and developed UI/UX interfaces for an AR/VR educational application for school children across multiple grade levels, enhancing engagement through animated and 3D interactive content.
- Wrote and maintained C# scripts within Unity 3D to implement scene management, interactive learning modules, and real-time 3D object behavior.
- Integrated AR/VR workflows into the Unity pipeline to deliver immersive educational experiences with 3D animated curriculum-aligned content.
- Collaborated on UI/UX design iterations in Figma and implemented responsive, child-friendly interfaces optimized for AR/VR headset and mobile AR platforms.

Intern – Indian Space Lab (ISL), Remote Dec 2023 – Jan 2024

- Modified flight controller and mission-planning code, improving UAV test-flight stability and navigation accuracy by ~30% through iterative testing and parameter optimization.
- Resolved a critical sensor calibration error affecting all remote sensing data, improving analysis accuracy by 35–40%.
- Enhanced CubeSat module sensor calibration, achieving a 15% improvement in data accuracy and 8% reduction in data noise during critical environmental testing.

PROJECTS

AI-Powered Quiz Generator | Java, Python

- Developed a Java Spring Boot application accepting PDF inputs and generating quizzes via an integrated Python ML pipeline using PyPDF2, Granite 3.2, Hugging Face transformers, and NLP techniques.
- Containerized the system with Docker achieving 90% question relevance accuracy.

Procedural Dungeon Generator | Python

- Designed a multi-staged hybrid procedural dungeon generator integrating rectangular rooms, handcrafted prefabs, and BFS cave expansion, achieving 100% connectivity across 100 test runs.
- Implemented evaluation metrics and visualization tools (CSV analytics, plots, animated GIF/MP4) to quantify path length (~45 tiles) and room diversity; built interactive Jupyter Notebook demo with sliders.

Verterra: Your World, Your Impact | Unity 3D, C#, AR/VR

- Developed an educational XR simulation in Unity 3D where players manipulate climate parameters (emissions, deforestation) across global regions to observe environmental impact.
- Implemented AR-based region placement mechanics inspired by Google Earth, enabling players to interactively adjust climate factors and restore ecosystems in their physical AR space.
- Designed a global balance system tracking interconnected environmental indicators, delivering a proof-of-concept for XR in climate education suitable for classroom and outreach environments.

CO-CURRICULARS

Lead, Social Media Team – VRGC, VIT Bhopal University (Jul 2023 – Ongoing) **Cyber-Security Awareness Program** – 5-day training by TheCyberDelta, Hackersvilla & GeeksForGeeks

HOBBIES & LANGUAGES

Hobbies: Tech Trendy, Gaming, Cooking, Classical Singing & Guitar, Photography **Languages:** Bengali, English, Hindi